

Operating system

Ch # 04

Objectives of Scheduling

- maximize CPU utilization
- maximize resource usage
- Maximize throughput → Completed no. of tasks in a unit time
- Minimize waiting time
- " response time
- " Turn around time
- Fairness
- Age-ing
- Uniformity
- CPU I/O burst time
 - ↳ CPU bound In which ^{process} CPU takes too much time in CPU than I/O
 - ↳ I/O bound In which process takes too much time in I/O than CPU
- Timer
- Dispatcher An algo. that responsible for context-switching

Times:-

- Arrival → when a process arrive
- Response → when CPU first time get the process
- Waiting → when a process wait to get CPU for some reason
- Burst
- Turnaround
- Completion

Algo Type:-

- Preemptive :

Can be interrupt

- Non-preemptive

Algo:-

(no interrupt)

FCFS : First Come First serve (Non-preemptive)

Process:	Arrival:	Burst:
P ₁	0	5
P ₂	3	3
P ₃	1	2
P ₄	2	1



← Gantt chart

CT:	Waiting T:	Turnaround: (CT - Arrival) ⇒ (WT + BT)
5	0	5
20	14	17
7	4	6
17	5	15

Response:

0

14

4

5

$$\text{Average waiting Time} = \frac{0 + 14 + 4 + 5}{4}$$

$$= 5.75$$